Recitation: Linear Functions Review Solutions

Problem 1. Suppose $f(x)$ and $g(x)$ are the functions given below.

$$f(x) = 2x - 3$$

$$g(x) = x^2 + 2x - 1$$

Compute the following:

(a) $f(5) = 2(5) - 3 = 10 - 3 = 7$

(b) $g(-2) = (-2)^2 + 2(-2) - 1 = 4 - 4 - 1 = -1$

(c) $f(0) - 3g(2) = (2 \cdot 0 - 3) - 3\big(2^2 + 2(2) - 1\big) = -3 - 3(7) = -3 - 21 = -24$

(d) $(f - g)(x) = f(x) - g(x) = (2x - 3) - (x^2 + 2x - 1) = 2x - 3 - x^2 - 2x + 1 = -x^2 - 2$

(e) $(fg)(x) = f(x)\,g(x) = (2x-3)(x^2+2x-1) = 2x^3+4x^2-2x-3x^2-6x+3 = 2x^3+x^2-8x+3$

(f) $\left(\dfrac{f}{g}\right)(x) = \dfrac{f(x)}{g(x)} = \dfrac{2x-3}{x^2+2x-1}$

(g) $(f \circ g)(0) = f\big(g(0)\big) = f(0^2 + 2(0) - 1) = f(0 + 0 - 1) = f(-1) = 2(-1) - 3 = -2 - 3 = -5$

(h) $(g \circ f)(0) = g\big(f(0)\big) = g(2(0) - 3) = g(0 - 3) = g(-3) = (-3)^2 + 2(-3) - 1 = 9 - 6 - 1 = 2$

(i) $(f \circ g)(x) = f\big(g(x)\big) = f(x^2 + 2x - 1) = 2(x^2 + 2x - 1) - 3 = 2x^2 + 4x - 2 - 3 = 2x^2 + 4x - 5$

(j) $(g \circ f)(x) = g\big(f(x)\big) = g(2x-3) = (2x-3)^2 + 2(2x-3) - 1 = (4x^2 - 12x + 9) + (4x - 6) - 1 = 4x^2 - 8x + 2$

Problem 2. Find the average rate of change for $f(x) = \dfrac{4-5x}{3}$ for $-1 \leq x \leq 3$ *without* performing any calculations. Then, find the average rate of change for $g(x) = x^2 - 4x + 5$ on the interval $[-1, 2]$.

We know that $f(x) = \dfrac{4-5x}{3} = \frac{4}{3} - \frac{5}{3}x$ is a linear function because it has the form $mx + b$ with $m = -\frac{5}{3}$ and $b = \frac{4}{3}$. We know that linear functions have a constant average rate of change, i.e. they have the same average rate of change on every interval, and that constant average rate of change is the slope, m. Therefore, the average rate of change of $f(x)$ on $-1 \leq x \leq 3$ is $m = -\frac{5}{3}$.

We have $g(-1) = (-1)^2 - 4(-1) + 5 = 1 + 4 + 5 = 10$ and $g(2) = 2^2 - 4(2) + 5 = 4 - 8 + 5 = 1$. But then the average rate of change for $g(x)$ on $[-1, 2]$ is…

$$\textit{Average ROC} = \frac{g(2) - g(-1)}{2 - (-1)} = \frac{1 - 10}{2 + 1} = \frac{-9}{3} = -3$$

Problem 3. Suppose that $h(x) = \sqrt{x-9} + 5$. Find at least two possible combinations for $f(x)$ and $g(x)$ such that $h(x) = (f \circ g)(x)$. Find the domain and range for $h(x)$.

There are infinitely many possible combinations for $f(x)$ and $g(x)$ (can you see why?). The two 'best' possible choices would be $g(x) = x - 9$, $f(x) = \sqrt{x} + 5$, and $g(x) = \sqrt{x-9}$, $f(x) = x + 5$. Now $h(x) = \sqrt{x-9} + 5$. We know that $x - 9$ is defined for any real number. Square roots are only defined if their inputs are nonnegative, i.e. greater than or equal to 0. So, we need $x - 9 \geq 0$, i.e. $x \geq 9$. So, the domain is $[9, \infty)$. We know that square roots produce nonnegative numbers, i.e. greater than or equal to 0. Once this number is added to 5, we obtain numbers greater than or equal to 5, i.e. $[5, \infty)$. Therefore, we have…

Domain: $[9, \infty)$

Range: $[5, \infty)$

Problem 4. Let $f(x)$ be the function given by $f(x) = 3x - 7$.

(a) Find a value in the range of f. Be sure to justify why the value is in the range.

(b) Compute $f(4)$. Is $(4, 1)$ on the graph of f? Explain.

(c) Is there an x such that $f(x) = 11$? Explain.

(d) Is $1 \in f^{-1}(3)$? Explain.

(e) Assuming f^{-1} exists, what is $f(f^{-1}(\pi))$ and $f^{-1}(f(\sqrt{2}))$?

Solution.

(a) *We know that the range of f is the set of outputs of f. Therefore, we can obtain an output by evaluating f at any value in its domain. For example, $f(0) = 3(0) - 7 = -7$, $f(10) = 3(10) - 7 = 23$, and $f(-5) = 3(-5) - 7 = -22$ are all values in the range of f.*

(b) *We have $f(4) = 3(4) - 7 = 12 - 7 = 5$. This implies that $(4, 5)$ is a point on the graph. Therefore, $(4, 1)$ cannot be on the graph of f. If it were on the graph, then we would know that $f(4) = 1$. But we know $f(4) = 5 \neq 1$.*

(c) *If there were x such that $f(x) = 11$, then...*

$$\begin{aligned} f(x) &= 11 \\ 3x - 7 &= 11 \\ 3x &= 18 \\ x &= 6 \end{aligned}$$

Of course, this assumes there is an x such that $f(x) = 11$; that is, we have shown that $x = 6$ is the only possible value. We can verify this possible solution: $f(6) = 3(6) - 7 = 18 - 7 = 11$. Therefore, there is such an x-value—namely, $x = 6$.

(d) *If $1 \in f^{-1}(3)$, then $f(1) = 3$. We have $f(1) = 3(1) - 7 = 3 - 7 = -4$. Therefore, $1 \notin f^{-1}(3)$.*

(e) *If f^{-1} exists, then we know that $(f \circ f^{-1})(x) = f\big(f^{-1}(x)\big)$ and $(f^{-1} \circ f)(x) = f^{-1}\big(f(x)\big)$ for all x. But then we would have $f(f^{-1}(\pi)) = \pi$ and $f^{-1}(f(\sqrt{2})) = \sqrt{2}$.*